HDOC DAY-14 REPORT

Menu-Driven C Program Using Switch-Case

Activity	HDOC Day-14
Batch	15 & 16
Language	C
Topic	Digit Uniqueness and Digit Frequency

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1. Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Check whether all digits of the number are unique.

2. Find the frequency of each digit from 0 to 9.

Given Activity

HDOC Day-14 Summarize this email

Dear Student,
The following are the **HDOC Day-14** activities for Batches 15 & 16:

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.
Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all its digits are unique.
  - **E.g. 1:** Number: 12345, Output: Unique
  - **E.g. 2:** Number: 12234, Output: Not Unique
- Find the frequency of each digit from `0` to `9`.
  - **E.g.** Number: 112333, freq_0 = 0, freq_1 = 2, freq_2 = 1, freq_3 = 3, freq_4 = 0, freq_5 = 0, freq_6 = 0, freq_7 = 0, freq_8 = 0, freq_9 = 0

Warm Regards,

2. Algorithm

Step 1: Start.

Step 2: Display a menu with two choices: (1) check unique digits and (2) find digit frequencies.

Step 3: Read the user's choice.

Step 4: Read a positive integer *n*.

Step 5: Use a switch statement on the selected choice.

Step 6 - Case 1: Initialize freq[0..9] to 0. Extract each digit using n % 10, increase its frequency, and remove the digit using integer division by 10. If any frequency is greater than 1, print "Not Unique"; otherwise print "Unique".

Step 7 - Case 2: Initialize freq[0..9] to 0. Extract each digit and count it. Display the frequency of every digit from 0 through 9.

Step 8 - Default: If the choice is neither 1 nor 2, display "Invalid choice!".

Step 9: Stop.

3. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Choice = 1 Number = 12345	Unique	Unique	Pass
2	Choice = 1 Number = 12234	Not Unique	Not Unique	Pass
3	Choice = 2 Number = 112333	freq_0=0, freq_1=2, freq_2=1, freq_3=3, freq_4...freq_9=0	freq_0=0, freq_1=2, freq_2=1, freq_3=3, freq_4...freq_9=0	Pass
4	Choice = 12234 Number = 12234	Invalid choice!	Invalid choice!	Pass

4. Program Code

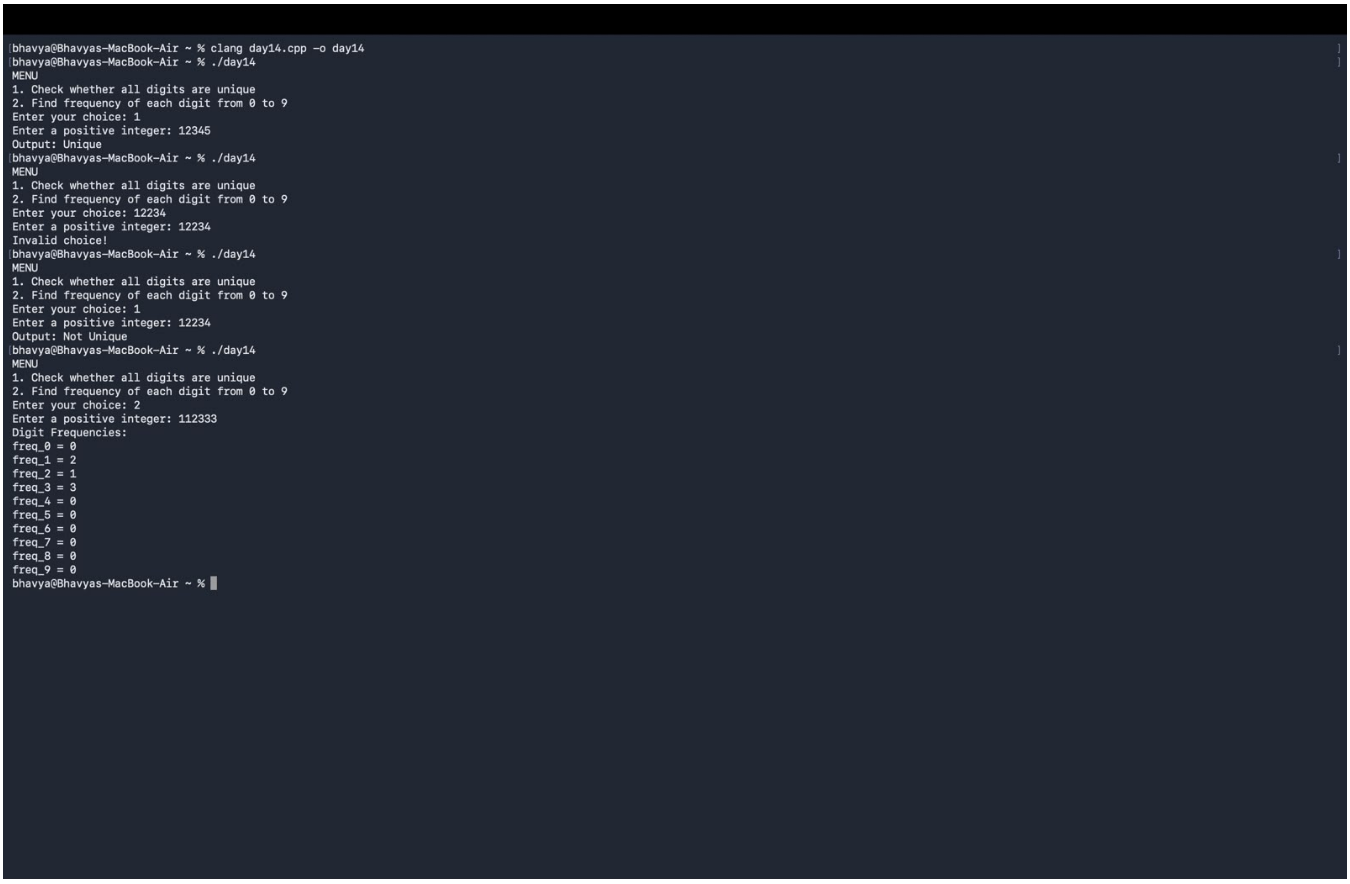
The following screenshots show the C program written for the Day-14 activity.

```
UW PICO 5.09                                  File: day14.cpp

#include <stdio.h>

int main()
{
    int choice, n, temp, digit;
    int freq[10];
    int i, unique;

    printf("MENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit from 0 to 9\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    switch (choice)
    {
        case 1:
            for (i = 0; i < 10; i++)
                freq[i] = 0;

            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            unique = 1;

            for (i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }

            if (unique == 1)
                printf("Output: Unique\n");
            else
                printf("Output: Not Unique\n");

            break;

        case 2:
            for (i = 0; i < 10; i++)
                freq[i] = 0;

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Pg       ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where is      ^V Next Pg       ^U UnCut Text    ^T To Spell
```

```
UW PICO 5.09                                  File: day14.cpp

                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            unique = 1;

            for (i = 0; i < 10; i++)
            {
                if (freq[i] > 1)
                {
                    unique = 0;
                    break;
                }
            }

            if (unique == 1)
                printf("Output: Unique\n");
            else
                printf("Output: Not Unique\n");

            break;

        case 2:
            for (i = 0; i < 10; i++)
                freq[i] = 0;

            temp = n;

            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            printf("Digit Frequencies:\n");

            for (i = 0; i < 10; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }

            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}

^G Get Help      ^O WriteOut      ^R Read File     ^Y Prev Pg       ^K Cut Text      ^C Cur Pos
^X Exit          ^J Justify       ^W Where is      ^V Next Pg       ^U UnCut Text    ^T To Spell
```

5. Compilation, Execution and Testing

The program was compiled and executed from the terminal. The test runs verify unique digits, repeated digits, digit frequencies, and invalid menu choice handling.



6. Result

The menu-driven C program was successfully compiled, executed, and tested. It correctly checks whether all digits are unique and calculates the frequency of each digit from 0 to 9 using switch-case.